

Abiel Kim

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TECHNICAL SKILLS

Programming: Python, Java, Go, C++, C#, Scala, Kotlin, PowerShell, Assembly (x86/ARM)
AI/ML: PyTorch, TensorFlow, scikit-learn, Feature Engineering, Model Training & Deployment
Backend: Node.js, Django, Spring Boot, Express.js, ASP.NET Core, Kafka, PostgreSQL, MongoDB, MySQL
Frontend: Next.js, React.js, TypeScript, JavaScript, Vue.js, HTML, Tailwind, CSS, Three.js, GraphQL
Cloud & DevOps: AWS (ECS, S3, Lambda, RDS), Docker, Kubernetes, Azure, GCP, Git, CI/CD Pipelines

EDUCATION

Simon Fraser University Burnaby, Canada
Bachelor of Science in Computing Science, Artificial Intelligence Specialization May 2025

EXPERIENCE

Machine Learning Engineer – MICA Technologies February 2025 – September 2025
Vancouver, BC Remote

- Achieved **production-grade** mIoU classification accuracies in an **accelerated** time frame
- Collaborated with geologists on **thin-section grain analyses** and inter-domain requirements
- Researched and developed proprietary architectures and **data processing** techniques
- Elicited client requirements to identify **workflow bottlenecks** and technical insights

Software Engineer – 3AG Systems May 2022 – August 2022
Burnaby, BC Hybrid

- Implemented UI dashboards in **Angular** and developed **backend APIs** in Node.js
- Built **interactive data analytics panels** with backend APIs to enable dynamic data rendering
- Handled **backend services** with AWS cloud infrastructure using REST APIs
- Debugged client and server end bugs to **optimize performance** and satisfy UX requirements

Product Engineer & Project Manager – Arlo Technologies September 2021 – April 2022
Richmond, BC On-site

- Verified **firmware builds** and soldered **PCB circuits** of both internal and external products
- Assisted in **program oversight**, logistics operations, and delegation of international tasks
- Executed **campus-wide** product experiments and **numerical analyses** to make internal decisions
- Evaluated, debugged, and resolved **defect products** to discover **root causes** in FW and HW

PROJECTS

SATMAP-7 | Zustand, Redis, Three.js, R3F, Drei, SGP4, GLSL, TypeScript, Tailwind 2026
Live Satellite Tracker 🌐 🔄

- Tracked thousands of satellites in real-time via precise **SGP4 propagation** on **CelesTrak GP TLE data**
- Rendered interactive point clouds via **R3F** over a **3D globe** with **GLSL day/night texture shaders**
- Architected a **triple-layer Redis caching pipeline** to **optimize API ingestion** and server performance
- Engineered **satellite search** with a smooth **camera interpolation animation** upon selection via **Drei**
- Designed both **mobile and desktop only HUD UIs** for navigation, statistics or **orbit path presentation**

OSSAD-9 | Zod, Zustand, Recharts, nmap, Firebase, TypeScript, Tailwind 2026
Computer Networking Toolbox 🌐 🔄

- Architected a **7-toolkit / 30+ sub-tool dashboard** with **WCAG, SOC 2, and RFC compliance**
- Implemented a colour-coded **network visualization library** from scratch for **LAN star topologies**
- Spawned **nmap** ping scan instances for a given **CIDR** with **SSE event streaming and logging**
- Traced **hop-by-hop network paths** with **TTL, IP, ASN, and RTT statistics** and **sparklines**
- Diagnosed **MITM attacks** via **ARP conflict detection** using **MAC-to-IP, OUI, and interface data**

Crystal Clear | *MongoDB, Mongoose, Google OAuth, Clouinary, JavaScript* 2026

Social Media Platform 🌐 🗣️

- Implemented a custom, real-time, **user messaging system** using MongoDB and Javascript
- Engineered a **comment section** UI and backend with **reply-threads** and comment-likes capabilities
- Built an asynchronous **search engine** with server integration and **query-prediction** capabilities
- Developed and tested my own user-personalized **content-recommendation** algorithms and backend
- Optimized server overloads via **paginated data fetches** and **client-server synchronization**

YouTube Generative AI | *HuggingFace, Python, Scikit, Matplotlib* 2025

SFU CMPT 713 Natural Language Processing 🗣️

- Designated **team lead** who designed our replica of YouTube's summary **generative AI** model
- Finetuned custom **Meta BART** and **Google T5** text-to-text transformers using **Kaggle** and **Python**
- Projected test data onto a 768D **sentence-embedding** space to identify **overlapping clusters**
- Optimized the cluster distribution via predefined K, **WCSS/Elbow Method**, and **Silhouette Scores**
- Experimented on and evaluated our generative models via **ROUGE n-gram overlap**

Brain-Computer Interface | *OpenBCI, Tensorflow, Python* 2021

AI-powered telepathy 🎮 🗣️

- Extracted **EEG brain waves** using the mark-IV **OpenBCI** 8-channel headset and Cyton board kit
- Trained neural networks to identify **motor-action intent** in the form of EEG (electroencephalography)
- Applied signal **processing theory** to compute max frequency rate for FFT EEG data streams
- Architected and programmed a **game engine and game loop** for **real-time visual feedback**
- Integrated the headset interface and inference model into the game loop to enable operability
- Demonstrated the capability to “**telepathically**” play the legendary game “Flappy Bird”

PUBLICATIONS AND RESEARCH

A Special Case of Nonlinear Extrapolation under the NTK December 2025

arXiv:2512.15749 🌐

- Pioneered a **theorem** which discovers when **NTK ReLU MLPs nonlinearly extrapolate** at the origin
- Proved 1 remark, 2 lemmas, and the main theorem by **pseudo-inversion** of the **NTK Gram matrix**
- Contributed nuance to the scientific community on how wide neural networks behave

A Data Scaling Law of a Manifold's Resolution April 2025

SFU CMPT 419 Special Topics in AI 🗣️

- Theorized a **neural scaling law** on the power-law of data scaling and neural network loss
- Earned an A+ with feedback from academics at the University of Illinois Urbana-Champaign
- Recognized for **exceeding expectations** and delivering one of the top submissions in Spring 2025

HONORS AND AWARDS

3x Dean's Honour Roll Spring 2021, Fall 2024 – Spring 2025

Recognized for outstanding academic excellence at SFU

4x Model United Nations October 2016 – March 2017

2nd place out of 1,000+ competitors at both VMUN and CAIMUN among others

World Scholar's Cup April 2017

2nd place out of 450 competitors at WSC regionals for writing

UBC MiniEnterprize February 2017

Top 4 finalist out of 500 competitors at UBC MiniEnterprize

LEADERSHIP AND INVOLVEMENT

Founder and President September 2015 – June 2018

Carson Graham Programming Club

Vice President of IT March 2017 – June 2017

Wish Youth Network Society